

Global potential for natural regeneration in deforested tropical regions



ICML



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Summary

This paper models the pantropical potential for natural forest regeneration at 30 m resolution by training a random forest classifier on 5.4 Mha of observed natural regrowth occurrences (2000–2016) and a comprehensive set of geophysical and socioeconomic covariates, ultimately using biophysical predictors for mapping. The authors estimate 215 Mha in tropical forest biomes has potential to naturally regenerate by 2030, with an above-ground carbon sequestration potential of 23.4 Gt C over 30 years, and identify strong spatial concentration in a handful of countries (Brazil, Indonesia, China, Mexico, Colombia). The work releases 30 m continuous probability layers and provides policy-relevant analysis for scaling cost-effective, nature-based climate solutions.



Strengths

Technical novelty and innovation

- Uses observed pantropical natural regrowth (distinct from plantations) to learn fine-scale drivers of regeneration potential, avoiding expert-opinion or potential-vegetation assumptions common in prior work.
- Integrates multi-scale biophysical features (forest proximity, soils, climate PCs, slope, NPP, fire, hydrology) and evaluates socioeconomic features, ultimately selecting a biophysical-only model for prospective mapping.
- Produces 30 m wall-to-wall probabilities conducive to operational restoration planning and optimization.

Experimental rigor and validation

- Very large training/validation sets; sensible feature engineering for remote sensing ecology.
- Variable selection and partial dependence analyses show coherent relationships (e.g., strong role of forest proximity and local forest density).
- Autocorrelation-aware validation analysis acknowledges spatial dependence.

Clarity of presentation

- Clear articulation of study scope, caveats (e.g., input resolution heterogeneity, underuse of natural regeneration), and use cases.
- Provides publicly accessible 30 m products via Zenodo and transparent data sources.

Significance of contributions

- High-impact, policy-relevant quantification of restoration potential and carbon removals with clear national and biome-level implications.
- Helps re-balance attention toward assisted natural regeneration as a scalable, lower-cost pathway, complementing tree planting.



Weaknesses

Technical limitations or concerns

- Probability calibration: Training and validation are balanced (1:1) but probabilities are interpreted as expected area and used to derive global totals; without prior correction or calibration, these are not posterior probabilities under true prevalence and can bias area estimates.
- Spatial validation: Random validation with post hoc distance-binning is weaker than spatial block cross-validation; reported “accuracy ~88%” on balanced samples likely overestimates generalization, especially in data-rich neighborhoods of training points.
- Label bias: Regrowth labels require persistence to 2016 and minimum height, inducing selection toward successful/mature regrowth and omission in humid biomes; this can bias model learning and under-represent conditions where early or transient regrowth occurs.

- Stationarity: Future potential (2030) assumes 2000–2016 conditions; excludes dynamic drivers (e.g., new roads, policy, commodity cycles, drought/fire trends).

Experimental gaps or methodological issues

- Limited baselines: No comparison to simpler or stronger learners (e.g., gradient boosting, LightGBM) or to parsimonious baselines (e.g., distance-to-forest-only) to bound achievable performance and assess incremental value of complex covariates.
- Metrics: Reliance on overall accuracy with balanced classes; lacks AUROC/AUPRC, calibration curves, Brier scores, or per-region error analysis tied to class imbalance and spatial heterogeneity.
- Uncertainty propagation: Carbon estimates adopt ranges from external products but do not propagate model and label uncertainty; extremely tight area CI (~0.44 Mha width) appears driven by pixel-counting rather than realistic error bounds.

Clarity or presentation issues

- The notion that “continuous potential can be interpreted as the probability of natural regeneration per pixel” is problematic without calibration; the manuscript should state this limitation prominently wherever probabilities are aggregated.
- Partial dependence plots are helpful but interpretations mix model marginal effects with mechanistic claims; caution warranted.
- Code availability “on request” falls short of best practices for reproducibility at ML venues.

Missing related work or comparisons

- Limited engagement with recent advances in spatially explicit modeling and spatial cross-validation practices, and with modern ML baselines used in remote sensing ecology (e.g., GBMs, calibration strategies).
- Minimal discussion of deep time-series and SAR-based approaches that might capture disturbance/recovery dynamics more robustly in persistently cloudy tropics.

Detailed Comments

Technical soundness evaluation

- The overall framework is well-motivated and the modeling choices are defensible for an applied, global-scale product. However:
 - Class balancing during training/validation inflates accuracy and distorts probability outputs. For area estimation and threshold-based sensitivity, prior correction or post-hoc calibration (e.g., Platt scaling or isotonic regression with true prevalence) is needed.
 - Spatial leakage is insufficiently controlled; spatial block cross-validation or leave-region-out strategies would yield more credible performance estimates and regional uncertainty bounds.
 - Label quality is uneven across biomes (notably humid forests), with known omission errors; the impact on learned patterns and predictions should be quantified (e.g., via sensitivity to noisy labels, geographically weighted error).
 - The argument for dropping socioeconomic variables (similar accuracy, higher stability) is practical, yet training on realized outcomes while excluding realized constraints risks conflating biophysical potential with historically realized potential; causal framing or sensitivity analyses would help.

Experimental evaluation assessment

- Positive: Large-scale, consistent processing; careful feature selection; coherent partial dependence; acknowledgment of regional accuracy differences.
- Improvements:
 - Report AUROC/AUPRC, calibration plots, and reliability diagrams, at least by biome/region.
 - Compare against gradient boosting (XGBoost/LightGBM) and a simple baseline (e.g., forest proximity + density) to contextualize gains.
 - Provide spatial block CV results (e.g., 50–100 km blocks), with maps of residuals/uncertainty to guide downstream use.
 - Quantify sensitivity to label omissions in the humid biome and to alternative definitions of “available for restoration.”

Comparison with related work (using the summaries provided)

- Relative to prior global restoration potential maps that rely on assumed potential vegetation or expert rules, this study’s use of empirical regrowth is a substantial advance in ecological realism.
- In contrast to recent ML/RS literature emphasizing deep temporal modeling (e.g., satellite image time series, SAR for deforestation monitoring) and ensemble learners for biomass estimation, the paper applies a conventional RF without exploring stronger modern baselines or temporal signals; discussing these tradeoffs (accuracy vs. interpretability and compute) would be valuable.
- The carbon quantification appropriately leverages Cook-Patton et al., but uncertainty from the regeneration model is not propagated; related work on design-based estimators and uncertainty accounting for forest change mapping suggests ways to strengthen inference.

Discussion of broader impact and significance

- The work is highly relevant to climate mitigation, biodiversity, and equity in restoration planning, and the data products will be useful for governments and NGOs. The manuscript commendably discusses leakage, permanence risks, policy alignment, and pitfalls of offsetting; it also cautions against misapplication in non-forest ecosystems.

- Given potential use in crediting, rigorous uncertainty and calibration are essential to prevent over-crediting and to prioritize assisted natural regeneration where permanence is plausible.

❓ Questions for Authors

1. How did you correct or calibrate predicted probabilities given the balanced training/validation? If not calibrated, can you quantify the bias this introduces to area totals and provide calibrated maps and revised totals?
2. Can you report AUROC/AUPRC and calibration metrics (e.g., Brier score, reliability plots) by biome/country, and provide spatial block cross-validation results to better control for spatial autocorrelation?
3. How sensitive are your results to the known omission errors in humid biome labels? Have you tried noise-robust training or label smoothing to mitigate these omissions?
4. Why was random forest chosen over gradient boosting (e.g., XGBoost, LightGBM) or modern spatial learners? Do such baselines materially change accuracy/calibration or variable importance?
5. The area CI is extremely narrow (~0.44 Mha). What uncertainty sources are included, and can you provide uncertainty that reflects label noise, model variance, and spatial autocorrelation?
6. Given that realized regeneration outcomes reflect socioeconomic constraints, how confident are you that a biophysical-only model trained on realized outcomes truly captures “biophysical potential”? Can you provide sensitivity analyses with and without socioeconomic predictors, and/or causal adjustments?
7. For carbon estimates, can you propagate the regeneration model’s uncertainty and offer country-level uncertainty envelopes? How sensitive are national totals to calibration and to the >0.5 threshold?
8. Will you release code and exact preprocessing scripts (including variable derivations and tiling) to enable full reproducibility and community extension?

✔ Overall Assessment

This is a timely and impactful contribution that leverages observed pantropical regrowth to produce the first 30 m, empirically grounded map of tropical natural regeneration potential and associated carbon benefits. The ecological and policy significance is high, and the data product will be valuable to planners and practitioners. From an ICML perspective, however, the methodological novelty is modest and several methodological aspects merit strengthening: probability calibration under class balancing, rigorous spatial cross-validation, comparisons with stronger modern baselines, and uncertainty propagation into carbon totals. The authors are transparent about many caveats, but the current evaluation framework may overstate accuracy and lead to overconfident area and carbon estimates. With the suggested improvements—especially probability calibration, spatially robust validation, stronger baselines, and full code release—this work would provide a more reliable foundation for high-stakes decision-making. As submitted, it is an excellent application study with substantial practical value but falls short of the methodological rigor and reproducibility standards typically expected at ICML.

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